



Multi-Mission Module Interface Control Document

DRAFT

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SIGNATURE TABLE

Position	Name	Signature
Government Representative		
Space Vehicle Representative		
M3 Representative		

ACRONYM LIST

Table 1. List of Acronyms

Abbreviation	Definition
C&T	Command and Telemetry Interface
CLA	Coupled Loads Analysis
ICD	Interface Control Document
MICD	Mechanical ICD (MICD M3 Enclosure XXXXXXXX)
MOI	Moment of Inertia
MPE	Maximum Predicted Environment
OAP	Orbit Average Power
OBCS	On-Board Computer System
TSP	Twisted Pair Impedance

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1. INTRODUCTION

This document details the interface requirements between the Host and the Multi-Mission Module (M3). “Host” is a reference to a Space Vehicle (SV) or the SV subsystem that interfaces with the M3. This document while maintained by SDA will be coordinated, prior to revision, with all contractors that are complying with it as governing document.

Many requirements come in pairs, written from the Host and M3 viewpoints, respectively. If conflicts are found between this Interface Control Document (ICD) and either a Host or M3 requirements document, this ICD takes precedence.

1.1 LOGICAL INTERFACES

At the highest level, all M3 hardware will be accommodated by each space vehicle provider’s bus design. The SV shall support M3s of various internal compute configurations that represent the M3’s capability when combined with the Secure Interoperable-middleware Layer (SIL) and mission applications. The M3 will interface to the Space Vehicle through wire harnesses that are provided by the space vehicle vendor (SVV).

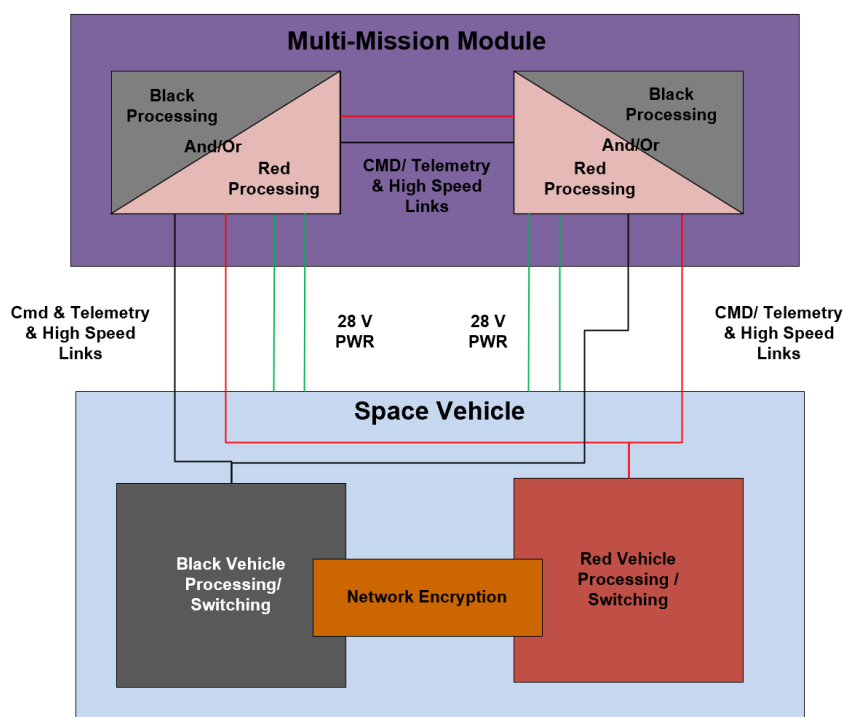


Figure 1. M3 Hardware Connectivity

As depicted in Figure 1, the processing capability of each M3 will cover the following configurations: (1) Dedicated Red processing, (2) Dedicated Black processing, (3) a split of Red and Black processing, and (4) a 50% footprint of these processing personalities.

2. M3 & HOST ICD REQUIREMENTS

2.1 MECHANICAL INTERFACE

2.1.1 MICD - Host

The host shall accommodate the M3 per the MICD.

2.1.2 MICD – M3

The M3 shall comply with the MICD.

2.1.3 M3 Mass Accommodation - Host

The host shall accommodate a M3 mass range of 2 kg up to 6.5 kg.

2.1.4 M3 Mass - M3

Total mass of the M3 shall be greater than 2 kg and less than or equal to 6.5 kg.

2.1.5 M3 Center of Mass - Host

The host shall accommodate the M3 center of mass range as depicted in the MICD.

2.1.6 M3 Center of Mass – M3

The M3 center of mass shall reside within the range depicted in the MICD.

2.1.7 Predicted M3 MOIs - Host

The host shall accommodate M3 principle MOIs range as depicted in the MICD.

2.1.8 Predicted M3 MOIs – M3

The predicted M3 principle MOIs shall be within the range depicted in the MICD.

2.1.9 M3 Venting

The M3 shall be designed for any enclosed volumes to have an ascent venting volume/area ratio of less than 2000 inches.

2.1.10 M3 Vibration Environment – M3

The M3 shall be tested to the following random vibration profile enveloping both the minimum workmanship and the maximum predicted environment (MPE).

NOTE: The environment is from GSFC-STD-7000 Rev A GEVS. The launch provider may update the MPE based on coupled loads analysis (CLA).

Table 2. Minimum Workmanship M3 Vibration Environment

Frequency (Hz)	PSD (g ² /Hz)
20	0.0053
150	0.04
800	0.04
2000	0.0064
	6.94 Grms

Table 3. Proto-qualification Level M3 Vibration Environment

Tx, 10.78 Grms	Tx, 10.78 Grms	Ty, 9.88 Grms	Ty, 9.88 Grms	Tz, 13.62 Grms	Tz, 13.62 Grms
Hz	G2/Hz	Hz	G2/Hz	Hz	G2/Hz
20	0.03	20	0.032	20	0.026
40	0.184	38	0.2	110	0.36
57	4.2	50	3.2	160	2.2
63	4.2	58	4	180	1.32
95	0.058	95	0.02	310	0.02
120	0.02	175	0.02	340	0.02
375	0.02	200	0.034	400	0.2
400	0.04	245	0.02	465	0.22
660	0.056	650	0.02	585	0.022
800	0.02	725	0.034	1000	0.02
1000	0.02	850	0.02	2000	0.005
2000	0.005	1000	0.02		
		2000	0.005		

2.1.11 M3 Vibration Environment - Host

The host's MPE shall not exceed the levels in the proto-qualification table of 2.1.10 at the interface between the bus and the M3.

2.2 ELECTRICAL / THERMAL INTERFACE

2.2.1 Input Voltage - Host

The host shall supply the M3 with 28V +/- 6V power sources. One set is intended to support red processing, and the other set is intended for black processing.

2.2.2 Input Voltage - M3

The M3 shall be capable of operating with one of the hosts provided power combinations.

2.2.3 Power Isolation – Host

One set of the host power inputs shall support red processing, and the other set shall support black processing. They shall be clearly identifiable and isolated from one another.

2.2.4 Power Isolation – M3

The M3 shall clearly identify which power input is red and which is black.

2.2.5 Data Connector - Host

The host data connectors for the M3 shall be PN MWDM2L-15SSB-186S connectors with a backshell, including jack screws. The wiring for this harness shall include 100-120 Ohm controlled impedance TSP. The pin out for the data connector sets are described in Host-M3 Data Connector Pinout below. The connectors to support red and black processing shall be uniquely identified.

2.2.6 Data Connector - M3

The M3 data connectors shall be PN MWDM2L-15PSB-186S connectors with a backshell, including jack posts. The wiring from the connector to printed circuit boards shall include 100-120 Ohm controlled impedance TSP. The M3 shall use any combination of the provided four connectors. The pinout is described below in Host-M3 Data Connectivity below.

Table 4. Host-M3 Data Connectivity

Connector Pinout					
Pin #	Signal Name	Twisted With	Wire	Input/Output	Interface Type
1	LVDS_Tx_P	2	100-120 Ohm TSP	Output	LVDS
2	LVDS_Tx_N	1	100-120 Ohm TSP	Output	LVDS
3	DGND	N/A	24-26 AWG	N/A	DGND (M3 inner shield)
4	PPS_P	5	100-120 Ohm TSP	Input	RS422
5	PPS_N	4	100-120 Ohm TSP	Input	RS422
6	DGND	N/A	24-26 AWG	N/A	DGND (Host inner shield)
7	10Mhz_Rx_P	8	100-120 Ohm TSP	Input	LVDS
8	10Mhz_Rx_N	7	100-120 Ohm TSP	Input	LVDS
9	DGND	N/A	24-26 AWG	N/A	DGND (Host inner shield)
10	CMD_422_P	11	100-120 Ohm TSP	Input	RS422
11	CMD_422_N	10	100-120 Ohm TSP	Input	RS422
12	DGND	N/A	24-26 AWG	N/A	DGND (Host inner shield)
13	TLM_422_P	14	100-120 Ohm TSP	Output	RS422
14	TLM_422_N	13	100-120 Ohm TSP	Output	RS422
15	DGND	N/A	24-26 AWG	N/A	DGND (M3 inner shield)
16	10 GbE	Internal	100-120 Ohm TSP	I/O	Ethernet
17	1.0 GbE_A	Internal	100-120 Ohm TSP	I/O	Ethernet
18	1.0 GbE_B	Internal	100-120 Ohm TSP	I/O	Ethernet
19	1.0 GbE_C	Internal	100-120 Ohm TSP	I/O	Ethernet

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2.2.7 Power Connector - Host

The host power connectors for the M3 shall be PN MWDM2L-15PSB-186S connector with a backshell, including jack screws. The pinout is described in Host-M3 Power Connector Pinout below.

2.2.8 Power Connector - M3

The M3 power connector shall be PN MWDM2L-15SSB-186S connector (mating connector) with a backshell, including jack posts. The matching pin out is described in Host-M3 Power Connector Pinout below.

Table 5. Host-M3 Power Connector Pinout

Pin #	Signal Name	Twisted With	Wire	Interface Type
1	RTN	9	24 AWG	GND
2	RTN	10	24 AWG	GND
3	RTN	11	24 AWG	GND
4	RTN	12	24 AWG	GND
5	RTN	13	24 AWG	GND
6	RTN	14	24 AWG	GND
7	RTN	15	24 AWG	GND
8	RTN	N/A	24 AWG	GND
9	PWR	1	24 AWG	34V
10	PWR	2	24 AWG	34V
11	PWR	3	24 AWG	34V
12	PWR	4	24 AWG	34V
13	PWR	5	24 AWG	34V
14	PWR	6	24 AWG	34V
15	PWR	7	24 AWG	34V

2.2.9 Inrush Current - Host

The host shall not provide greater than 6A of M3 inrush current, the maximum inrush current duration shall not exceed 100 ms.

2.2.10 Inrush Current - M3

The M3 shall be compatible with a maximum inrush current of 6 A for a duration of up to 100 ms.

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2.2.11 Input Supply Capacitance - Host

The host shall accommodate a M3 input supply capacitance of at least 500 microFarads.

2.2.12 Input Supply Capacitance - M3

The M3 shall have no more than 500 microFarads of capacitance on the power supply input.

2.2.13 Command and Telemetry Interface - Host

The host shall communicate with the M3 via an RS422 C&T interface at a rate that is no lower than 231 kbps.

2.2.14 Cmd/Tlm RS422 Configuration - Host

The host RS422 configuration for the command and telemetry link shall use 1 start bit, 1 stop bit, and no parity.

2.2.15 Command and Telemetry Interface - M3

The M3 shall communicate with the host via a RS422 C&T interface at a rate greater than 231 kbps.

2.2.16 Cmd/Tlm RS422 Configuration - M3

The M3 RS422 configuration for the command and telemetry link shall use 1 start bit, 1 stop bit, and no parity.

2.2.17 Peak Power Accommodation - Host

The host shall provide a peak power to the M3 that accommodates its maximum power draw, up to a maximum of 200 W, with no constraints of when the power is provided.

2.2.18 Peak Power Limitation - M3

The M3 shall not exceed a peak power draw of more than 200 W and shall not exceed 100 W for a 50 % module configuration.

2.2.19 Timing - Host

The host shall provide a pulse per second (PPS) signal with the timing described in Figure 2 below.

2.2.20 Timing - M3

The M3 shall support a PPS signal with the timing described in Figure 2 below.

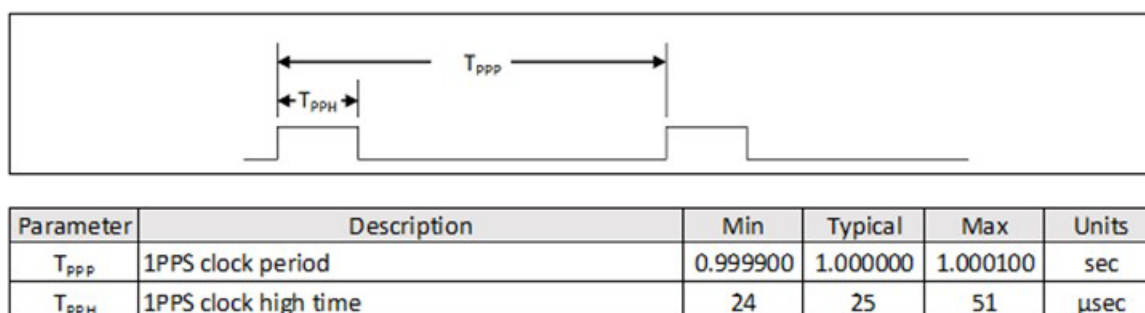


Figure 2. PPS Timing Diagram

2.2.21 PPS - Host

The host shall provide a PPS differential signal to the M3 at RS422 levels.

2.2.22 PPS - M3

The M3 shall support a PPS differential signal from the host at RS422 levels.

2.2.23 Bulk Data Interface – Host

The host shall communicate with the M3 via a RS422 Bulk Data interface at a minimum of 230.4 kbps.

2.2.24 Bulk Data RS422 Configuration - Host

The host RS422 configuration for the bulk data link shall use 1 start bit, 1 stop bit, and no parity.

2.2.25 Bulk Data Interface - M3

The M3 shall communicate with the host via a RS422 Bulk Data interface at a minimum of 230.4 kbps.

2.2.26 Bulk Data RS422 Configuration - M3

The M3 RS422 configuration for the bulk data link shall use 1 start bit, 1 stop bit, and no parity.

2.2.27 Command and Telemetry Compliance

The M3 and the SV will be compliant with message definition as described in the Command and Telemetry Dictionary (SDA-XXXXX).

2.2.28 Input Voltage Rise Time - Host

The host power switch to the M3 shall have a minimum rise time of 500 μ sec.

2.2.29 Input Voltage Rise Time - M3

The M3 shall accept an input voltage with a minimum rise time of 500 μ sec.

2.2.30 RS422 Termination - Host

The host shall terminate the bulk data, C&T, and PPS RS422 circuits per Figure 3 and Figure 4 below.

2.2.31 RS422 Termination - M3

The M3 shall terminate the bulk data, C&T, and PPS RS422 circuits per Figure 3 and Figure 4 below.

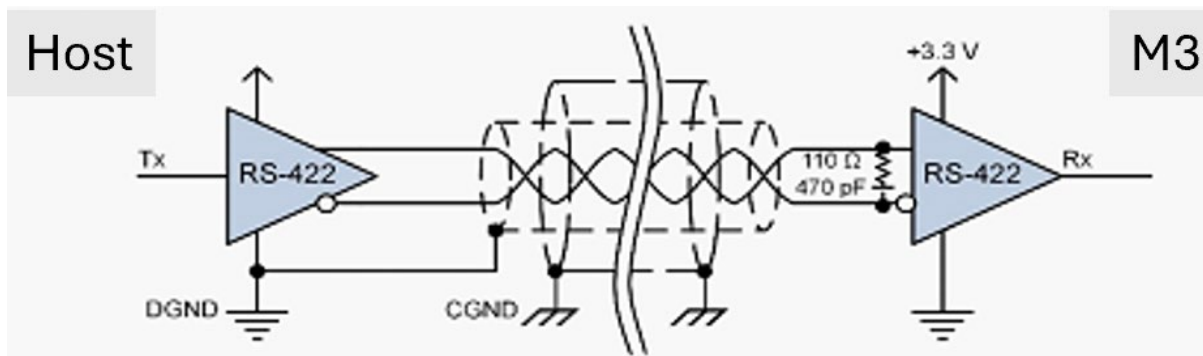


Figure 3. RS422 Interface Diagram from Host to M3

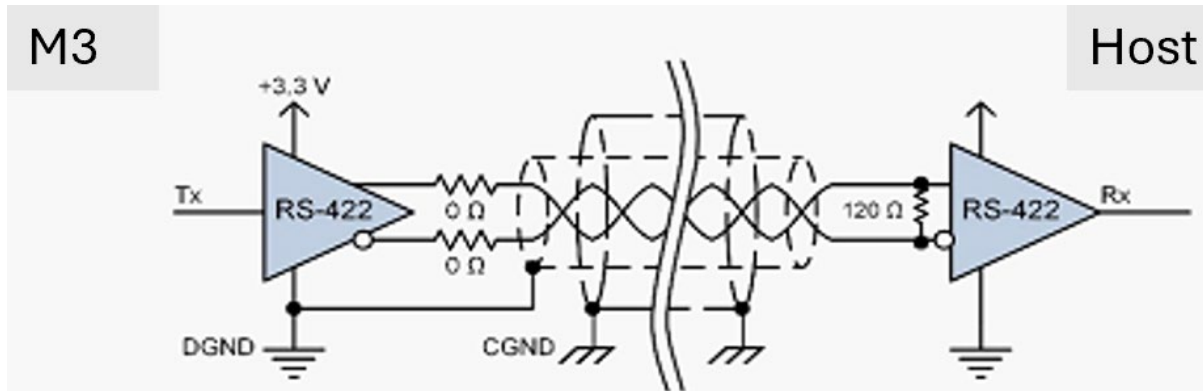


Figure 4. RS422 Interface Diagram from M3 to Host

2.2.32 PPS Time Stamp - Host

The host shall provide the PPS time stamp via the C&T RS422 interface.

2.2.33 PPS Time Stamp - M3

The M3 shall receive the PPS time stamp via the C&T RS422 interface.

2.2.34 Graceful Payload Shut Down - Host

The host shall provide a signal to the M3 3 seconds prior to turning the M3 off.

2.2.35 Graceful M3 Shut Down - M3

The M3 shall safely shut down within 3 seconds after receiving the host signal to shut down.

2.2.36 Real-time Transfer Data Rate - M3

The M3 shall provide configurable real time telemetry. The telemetry data rate shall be a minimum of 80 kbps and a maximum of 1 Mbps of real time telemetry.

2.2.37 Real-time Transfer Data Rate – Host

The host shall accommodate a minimum of 80 kbps and a maximum of 1 Mbps of real time telemetry from the M3.

2.2.38 Thermal Interface Temperature & Heat Transfer Performance – M3

The M3 shall meet all its operating modes with a thermal interface as provided by the host.

2.2.39 Thermal Interface Temperature & Heat Transfer Performance – Host

The SV thermal interface structure shall be maintained below 40 deg C and above 0 deg C with an interface heat/power transfer performance of $> 12\text{W/m-K}$. The thermal hosting environment of the M3 shall not introduce more than 5 W of thermal input exclusive of the interface plate.

2.3 COMMANDING REQUIREMENTS

2.3.1 Time at Next PPS

The host shall deliver a Time at Next PPS message to the M3 at least 1 ms after the previous PPS and more than 500 ms prior to the next PPS.

2.3.2 Data Endianness

All multi-byte fields transmitted between the M3 and the Host shall be in Big-endian byte order.

2.3.3 Command and Telemetry Interface Packetization

All data transmitted over the C&T RS-422 interface shall be encapsulated in CCSDS Space Packets in accordance with CCSDS 133.0-B-2 as shown in Figure 5 and Figure 6 below.

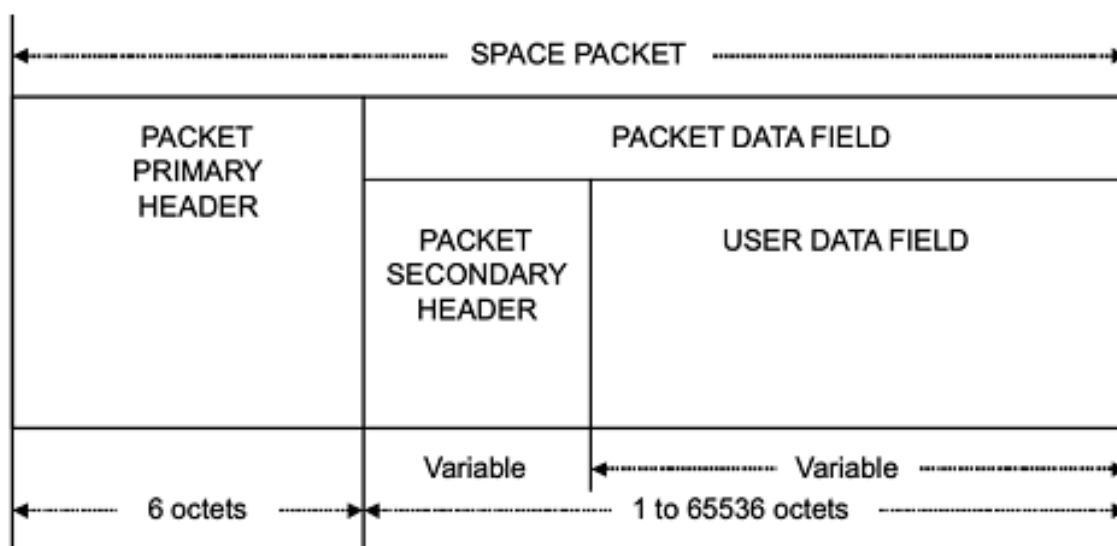


Figure 5. Space Packet Format - CCSDS 133.0-B-1 Figure 4-1.

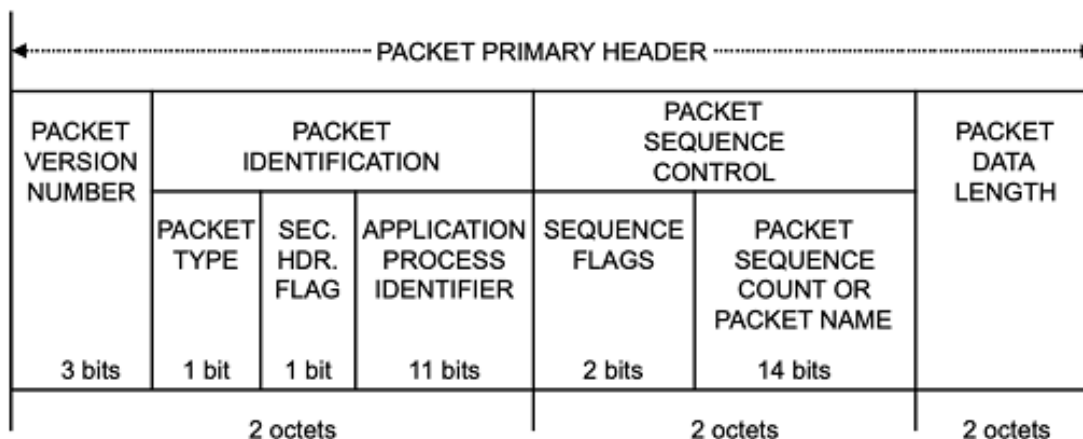


Figure 6. Space Packet Primary Header - CCSDS 133.0-B-1 Figure 4-2.

2.3.4 Space Packet Sequence Count

All space packets transmitted over the C&T interface shall populate the Packet Sequence Count as a running counter of packets sent over the interface.

2.3.5 Space Packet Secondary Header

All space packets transmitted over the C&T interface shall include a 9-byte secondary header in accordance with Figure 7 below.

NOTE: Field descriptions for the secondary header are described in Space Packet Secondary Header Field Descriptions below.

M3 Space Packet Secondary Header 9 Bytes					
Message Type UInt16	Source SCID UInt16	Source Sub Component UInt8	Destination SCID UInt16	Destination Sub Component UInt8	Streaming Flag UInt8

Figure 7. Space Packet Secondary Header

Table 6. Space Packet Message Types

Message Type	Message Name	Field Name	Data Type	Units	Description
Bidirectional Messaging (Either Host to M3 OR M3 to Host)					
0	ACK				An acknowledgement of a received space packet or message Every space packet must be acknowledged with either an ACK or NAK response. While each message needs acknowledged, in order to reduce message traffic, ACKs do not need to be acknowledged individually. The receipt of an ACK for space packet sequence "N" will assume the acceptance of all previous, unacknowledged packets N-1, N-2, ... etc. All NAK's must be immediately reported. The receipt of a NAK should be preceded with an ACK for any accepted, but unacknowledged, packets. Senders should maintain a minimum of 5 seconds worth of delivered packets to enable re-sending failed messages. Either an ACK or a NAK must be sent at a minimum of every 5 seconds to enable clearing the senders retry queues.
		SequenceCounter	u16		Sequence counter of the space packet being acknowledged
1	NAK				Negative acknowledgement of a received space packet (missing packet counter, failed checksum, etc.) See notes in ACK
		SequenceCounter	u16		Sequence counter of the space packet that was missing/corrupt/rejected. In case a segmented message is rejected, this should be the last sequence counter in the rejected message.

Message Type	Message Name	Field Name	Data Type	Units	Description																		
		Reason	u8	enum	Why the segment/message was rejected <table><tr><th>Value</th><th>Description</th></tr><tr><td>0</td><td>Corrupt/missing (segment re-request)</td></tr><tr><td>1</td><td>Invalid destination SCID</td></tr><tr><td>2</td><td>Invalid destination sub-component</td></tr><tr><td>3</td><td>Routing failure (e.g., crosslink not available)</td></tr><tr><td>4</td><td>Unrecognized message type</td></tr><tr><td>5</td><td>Incorrect size</td></tr><tr><td>6</td><td>Invalid parameter</td></tr><tr><td>255</td><td>Other reason</td></tr></table>	Value	Description	0	Corrupt/missing (segment re-request)	1	Invalid destination SCID	2	Invalid destination sub-component	3	Routing failure (e.g., crosslink not available)	4	Unrecognized message type	5	Incorrect size	6	Invalid parameter	255	Other reason
Value	Description																						
0	Corrupt/missing (segment re-request)																						
1	Invalid destination SCID																						
2	Invalid destination sub-component																						
3	Routing failure (e.g., crosslink not available)																						
4	Unrecognized message type																						
5	Incorrect size																						
6	Invalid parameter																						
255	Other reason																						
		DetailLength	u8	bytes	Length of the "Detail" field																		
		Detail	u8[]		ASCII string explaining "Reason" (e.g., "PosX out of bounds"). This may be empty.																		
2	Echo				A simple ping-style message for connectivity testing. May be used as an aliveness watchdog. Expected response is an "Echo Reply".																		
		MessageLength	u8	bytes	Number of bytes in the "Message" field																		
		Message	u8[]		An ASCII string to be echoed																		
3	Echo Reply				Response to an "Echo" message																		
		MessageLength	u8	bytes	Number of bytes in the "Message" field																		
		Message	u8[]		The ASCII string being echoed																		
Host to M3 Messaging																							

Message Type	Message Name	Field Name	Data Type	Units	Description									
100	Generic M3 Commanding				Commanding that is ground of M3-known, but for which the host doesn't need full definitions (e.g., mode transitions, software updates, etc.)									
		Data	u8[]		Data in a format known to the M3. Reference the M3 software data dictionary, SDA/XX-XXXX.									
101	Time at Next PPS				The time shall be delivered to M3 within a specified range as described in section 3.3.1.									
		Seconds	u32	seconds	The number of seconds since January 1 2000 12:00:00 terrestrial time (J2000 TT), synchronized with a PPS signal. The time shall represent the time at the NEXT received PPS.									
		Flags	u8		Bit field describing abnormal states about the upcoming time or PPS: <table><thead><tr><th>Bit</th><th>Name</th><th>Description</th></tr></thead><tbody><tr><td>0</td><td>Unreliable</td><td>The host time has not been initialized from a reliable external source (e.g., ground command, GNSS) yet, or it has not been refreshed for a significant period of time.</td></tr><tr><td>1</td><td>Initialized</td><td>The host time has just been (re-)initialized from a reliable external source. Expect time to jump.</td></tr></tbody></table>	Bit	Name	Description	0	Unreliable	The host time has not been initialized from a reliable external source (e.g., ground command, GNSS) yet, or it has not been refreshed for a significant period of time.	1	Initialized	The host time has just been (re-)initialized from a reliable external source. Expect time to jump.
Bit	Name	Description												
0	Unreliable	The host time has not been initialized from a reliable external source (e.g., ground command, GNSS) yet, or it has not been refreshed for a significant period of time.												
1	Initialized	The host time has just been (re-)initialized from a reliable external source. Expect time to jump.												
102	Host OBCS State				Storage capacity available for M3 file transfers, etc.									
		FileTransferCapacity	u64	bytes	Free space available for file M3-to-host file transfers									
103	Load Shed Request				Request M3 to gracefully load shed									
		(no fields)												
104	Shutdown Warning				Indicates the host will be shutting down the M3 and the M3 should begin "safing" itself.									

Message Type	Message Name	Field Name	Data Type	Units	Description
					Expected response is "Shutdown Ready" at which point it is safe to shutdown the M3.
M3 to Host Messaging					
200	Generic M3 Telemetry				Telemetry that is ground or M3-known, but for which the host doesn't need full definitions
		Data	u8[]		Data in a format known to the destination. Reference the M3 software data dictionary, SDA/XX-XXXX.
201	Shutdown Ready				Response to the host that M3 "safing" is complete and prepared for shutdown.
202	Power Cycle Request				Request M3 power to be cycled

Table 7. Space Packet Sub-Component Identifiers

ID	Description
1	Host's On-Board Compute System (OBCS)
2	M3

Table 8. Space Packet Secondary Header Field Descriptions

Field Name	Description
Message Type	A message-type identifier for the encapsulated message, defining the internal message parameters, as defined in Table 8.
Source SCID	The identifier for the spacecraft generating the message
Source Sub-Component	An identifier of the sub-component within the spacecraft generating the message, as defined in Table 8.
Destination SCID	The identifier for the spacecraft to whom the message is being delivered.
Destination Sub-Component	An identifier of the sub-component within the destination spacecraft, as defined in Table 8.
Streaming	This packet contains real-time telemetry that should be sent to the ground directly. It does not need to be archived by the host vehicle.

2.3.6 Message Interface Synchronization

All space packets transmitted between the M3 and the host, including both the CMD/TLM and Bulk Data Transfer interfaces, shall begin with a 2-byte synchronization marker of hexadecimal 0xBEEF in accordance with Figure 9.

2.3.7 Message Interface Checksum

All space packets transmitted between the M3 and the host, including both the CMD/TLM and Bulk Data Transfer interfaces, shall be terminated with a 4-byte CRC in accordance with Figure 8 below.

**Figure 8. Space Packet Synchronization and Validation**

2.3.8 Message Interface Checksum Algorithm

The Message Interface checksum field shall calculate a CRC-32c with algorithm parameters specified in the following table, across the entire Space Packet.

Table 9. CRC-CCITT Parameter Definitions

CRC-CCITT Parameter Name	Value
Polynomial	0x1edc6f41
Initial Value	0xFFFFFFFF
Input Reflection	Yes
Output Reflection	Yes
Output XOR	0xFFFFFFFF

2.3.9 M3 Space Packet Re-Request

The host shall be capable of re-transmitting a minimum of the most recent 20 Space Packets upon request from the M3.

2.3.10 Host Space Packet Re-Request

The M3 shall be capable of re-transmitting a minimum of the most recent 20 Space Packets upon request from the host.

2.3.11 File Transfers

All file transfers between the M3 and the host shall utilize the Bulk Data Transfer RS-422 interface.

2.3.12 Initiate Uploads with Specific Packet

File uploads from the host shall be initiated with a "Initialize File Uplink" packet as shown in the M3 data dictionary (SDA/XX-XXXX).

2.3.13 Begin with and Increment Uplink ID

For each "Initialize File Uplink" packet, the host shall begin with an UplinkID of 1 and increment the Uplink ID for each subsequent file upload.

2.3.14 Compress Files for Transfer

If transferring compressed files from host to M3, the host shall compress the complete file using gzip prior to transmission.

2.3.15 Reply to File Uplink Packet

The M3 shall reply to an "Initialize File Uplink" packet with a "Initialize File Uplink Reply" packet as shown in the M3 data dictionary (SDA/XX-XXXX).

2.3.16 Uplink File Contents Using Uplink ID

Following an "Initialize File Uplink" packet, the host shall send a sequence of "Uplink File Segment" packets containing the file's contents and an Uplink ID field matching the ID associated with the previously sent "Initialize File Uplink" packet.

2.3.17 Uplink File Segment Size

Each "File Segment Uplink" packet shall have a maximum file segment data size of 960 bytes.

2.3.18 Send Uplink Complete Packet

The M3 shall send a "File Uplink Complete" packet, as shown in the M3 data dictionary (SDA/XX-XXXX), to the host upon receipt of the final "Uplink File Segment".

2.3.19 Send Downlink Reply Packet

The M3 shall send a "Initialize File Downlink Reply" packet, as shown in the M3 data dictionary (SDA/XX-XXXX), to indicate the beginning of a file transfer to the host.

2.3.20 Downlink File Contents Using Downlink ID

After sending a "Initialize File Downlink Reply" packet, the M3 shall send a series of "File Segment Downlink" packets containing the file contents with an associated Downlink ID field matching the ID associated with the previously-sent "Initialize File Downlink Reply" packet.

2.3.21 Downlink File Segment Size

Each "File Segment Downlink" packet shall have a maximum file segment data size of 960 bytes.

2.3.22 Send Downlink Complete Packet

The M3 shall send a "Downlink Complete" packet, as shown in the M3 data dictionary (SDA/XX-XXXX), to the host upon sending the final "File Segment Downlink" packet.

2.3.23 Send Cancel Downlink

The host shall be capable of sending a "Cancel File Downlink" packet, as shown in the M3 data dictionary (SDA/XX-XXXX), to cancel a file transfer to the host.

2.3.24 Cancel Downlink after Allotted Time

The host shall be capable of canceling a file downlink if it has not received a file segment within 30 seconds of the last message received for an associated downlink ID.

2.3.25 Cancel Uplink after Allotted Time

The M3 shall be capable of canceling a file uplink if it has not received a file segment within 30 seconds of the last message received for an associated uplink ID.